Jonathan Drechsel

PhD Student in Computer Science · University of Passau, Germany

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EDUCATION

2024 – present	**Ph.D. in Computer Science**, University of Passau, Germany Advisor: Prof. Dr. Steffen Herbold · Chair of AI Engineering
2022 – 2023	**M.Sc. in Computer Science**, Dresden University of Technology **Grade: 1.1**
2021 – 2024	**M.Sc. in Mathematics**, Dresden University of Technology **Grade: 1.2**
2019 – 2021	**B.Sc. in Computer Science**, Dresden University of Technology **Grade: 1.6**
2017 – 2021	**B.Sc. in Mathematics**, Dresden University of Technology **Grade: 1.8**

German grading scale (1.0 is the best possible grade).

PUBLICATIONS

Drechsel, Jonathan and Steffen Herbold (2026). "The GRADIEND Python Package: An End-to-End System for Gradient-Based Feature Learning". In: *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP 2026)*.

Drechsel, Jonathan, Erisa Bytyqi, and Steffen Herbold (2026). "Understanding or Memorizing? A Case Study of German Definite Articles in Language Models". In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. **(Oral Presentation)**.

Drechsel, Jonathan and Steffen Herbold (2026). "GRADIEND: Feature Learning within Neural Networks Exemplified through Biases". In: *International Conference on Learning Representations (ICLR 2026)*.

Drechsel, Jonathan, Anja Reusch, and Steffen Herbold (2025). "MAMUT: A Novel Framework for Modifying Mathematical Formulas for the Generation of Specialized Datasets for Language Model Training". In: *Transactions on Machine Learning Research (TMLR)*.

Simon Birkenheuer, **Jonathan Drechsel**, Paul Justen, Jimmy Pöhlmann, Julius Gonsior, and Anja Reusch (2023). "Sabrina Spellman at SemEval-2023 Task 5: Discover the Shocking Truth Behind this Composite Approach to Clickbait Spoiling!" In: *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval 2023)*.

Lucas Woltmann, **Jonathan Drechsel**, Claudio Hartmann, and Wolfgang Lehner (2022). "Ingredient-based Forecast of Sold Dish Portions in Campus Canteen Kitchens". In: *2022 IEEE 38th International Conference on Data Engineering Workshops (ICDEW)*.

RESEARCH & WORK EXPERIENCE

2024 – present	**Research Assistant**, University of Passau, Germany • Research on mechanistic interpretability, targeted feature learning, and math NLP • Teaching assistant for undergraduate and graduate courses in Computer Science and AI Engineering
2022 – 2024	**Working Student (Software Development)**, Liofit GmbH
2020 – 2022	**Working Student (Software Development)**, Conimon GmbH

TEACHING

Teaching Assistant at the University of Passau (B.Sc. & M.Sc. programs in Computer Science and AI Engineering):

SS 2026	**Programming for Data Analytics**
WS 25/26, 26/27	**Introduction to AI Engineering**
SS 2025	**Deep Learning for Natural Language and Code**
SS 2025	**Requirements Engineering**
WS 2024/25	**Principles of AI Engineering**
SS 2024	**AI Engineering Lab**

TECHNICAL SKILLS

Programming: Python, Java (proficient); C, C++ (good knowledge)

Machine Learning: PyTorch, Hugging Face Transformers, NumPy, SciPy

Languages: German (native), English (fluent), French (basic)